



Amatria's 6th Birthday Celebration

Thursday, April 11

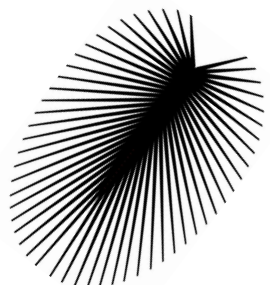
9 A.M.

In person at Luddy Hall 4th Floor under Amatria and via Zoom

Zoom Link : <https://iu.zoom.us/j/88426214568>

Event Schedule

9:00am	Welcome and brief introductions
9:10am	Talk by Philip Beesley
9:20am	Announcement of Martin-Bewley Amatria Support Fund by Ree Palmer Easton
9:25am	Musical Dance Performance by Sophia Long
9:30am	Tavola: A Visualization for Amatria by Andreas Bueckle
9:40am	Amrut & Ansh by David Ebbinghouse
9:50am	Cake and VR Demos
10:00am	End



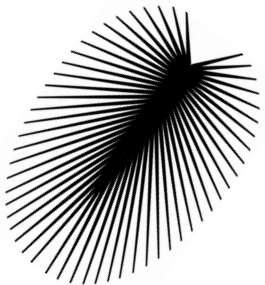
Katy Börner



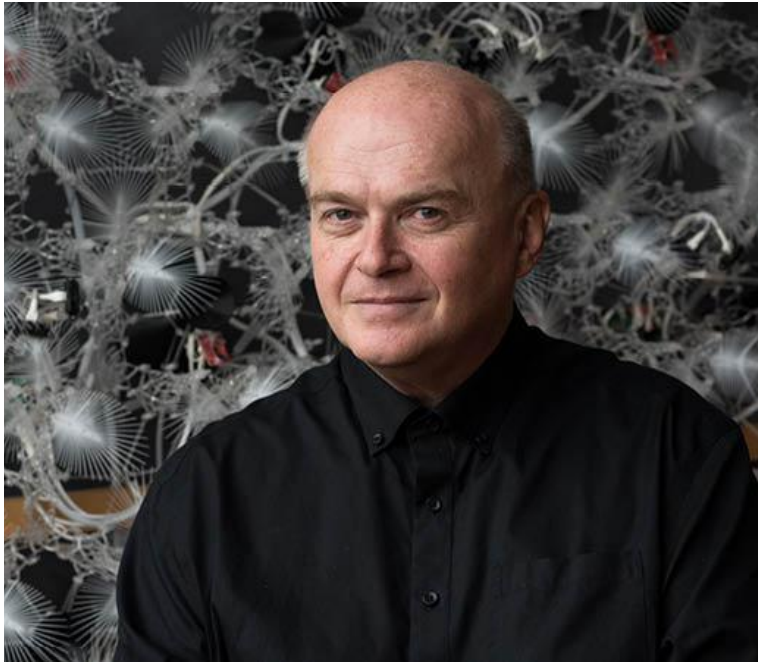
Katy Börner is the Victor H. Yngve Distinguished Professor of Engineering and Information Science in the Departments of Intelligent Systems Engineering and Information Science, Luddy School of Informatics, Computing, and Engineering; core faculty of the Cognitive Science Program; and founding director of the Cyberinfrastructure for Network Science Center (<http://cns.iu.edu>)—all at Indiana University in Bloomington, Indiana.

She is also a visiting professor at the Royal Netherlands Academy of Arts and Sciences (KNAW) in the Netherlands and Humboldt Fellow, Dresden University of Technology, Germany. Börner became a Fellow of the American Association for the Advancement of Science (AAAS) in 2012, a Humboldt Research Fellow in 2017, and an Association for Computing Machinery (ACM) Fellow in 2018. Since 2005, she serves as a curator of the international *Places & Spaces: Mapping Science* exhibit (<http://scimaps.org>).

She has very warm, motherly feelings toward Amatria and loves to see her grow up surrounded by much human intelligence and interaction.



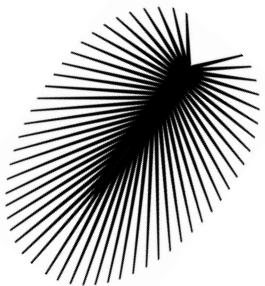
Philip Beesley



Philip Beesley is a practicing visual artist, architect, and Professor in Architecture at the University of Waterloo and Professor of Digital Design and Architecture & Urbanism at the European Graduate School. Beesley's work is widely cited in contemporary art and architecture, focused in the rapidly expanding technology and culture of responsive and interactive systems.

He serves as the Director for the Living Architecture Systems Group, and as Director for Riverside Architectural Press. His Toronto-based practice, Philip Beesley Architect Inc., operates in partnership with the Europe-based practice Pucher Seifert and the Waterloo-based Adaptive Systems Group, and in numerous collaborations including longstanding exchanges with couture designer Iris van Herpen and futurist Rachel Armstrong.

As an internationally recognized expert and pioneer in kinetic, responsive, near-living architectural installations, Philip Beesley leads the LASG Partnership and the Scaffolds research stream. As the leader of numerous collaborative projects, Beesley has senior level experience managing large, complex multidisciplinary teams and large-scale public architecture. Work resulting from the previous SSHRC partnership were presented in several public venues including the 2012 Biennale of Sydney. As a Professor at the University of Waterloo, School of Architecture he has mentored +1500 architecture students. He has authored and edited 19 books and has been the chair for numerous symposia and conferences.



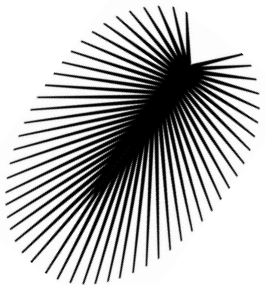
Andreas Bueckle



Andreas Bueckle (<https://andreas-bueckle.com>) is the Research Lead in the Luddy School of Informatics, Computing, and Engineering at Indiana University. Coming from a background in video journalism and digital media, Andreas researches data visualization and extended reality (XR) environments. He holds a B.A. in Media Studies from Eberhard Karls University in Tuebingen, an M.A. in Communications in Economic and Social Contexts from Berlin University of the Arts (Germany), and a Ph.D. in Information Science from Indiana University.

Andreas has been involved with Philip Beesley's work since 2016. In 2017, he developed a 3D visualization of light and motor activity within the *Sentient Veil*, another piece by Philip Beesley on exhibit at the Isabella Stewart Gardner Museum in Boston, MA. In 2018, he developed and deployed a similar 3D visualization under Amatria called *Tavola*, which is used by new and old visitors to Amatria. In 2019, he served as one of the keynote speakers (together with Katy) at the Living Architecture Symposium at OCAD University, Toronto (ON), Canada.

Besides his passion for XR technologies and software development, Andreas has a deep passion for the visual arts, specifically videography (in which he specialized during his undergraduate education) and landscape photography on six continents.



Sophia Long



Name: Sophia Long

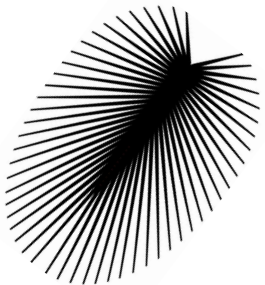
From: Sarasota, Florida

Year in School: Senior

Major: Ballet and Neuroscience Psychology

I have been dancing for 18 years of my life, and attended a professional ballet school in Boston Massachusetts for 3 years.

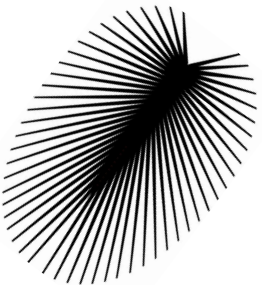
(photo is just a placeholder until we get a professional one)



David Ebbinghouse



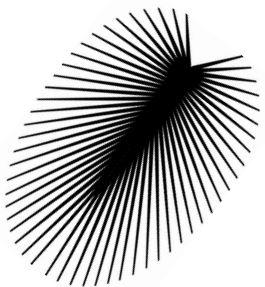
David Ebbinghouse is a practicing local artist. Since graduating from Indiana University with a degree in Fine Arts, he has found numerous ways to interact with the Bloomington Community. His creation of an alter ego, Tom Cat Spray, who anonymously left intricate stencil graffiti in the alleyways of Bloomington in the 1980's is probably his best known work. An example of this style of painting is on permanent display in Ballentine Hall. For the past eleven years he has designed three outfits each year for the Trashion-Refashion runway show held at the Buskirk Chumley Theatre. He has contributed articles to The Ryder Magazine periodically. The last series (July/August, September/October 2023 and March/April 2024) discussed the artistic legacy of former Sculpture Professor Jean-Paul Darriau, his public works around Bloomington, and Ebbinghouse's time as Darriau's Studio Assistant. The Grunwald Gallery in the Fine Arts Building gave Ebbinghouse a mid career retrospective "Core Samples" in 1995.



In the past 6 years, *Amatria* had lots of guests, visitors, friends around her.

She had many big tours, including one for Indiana Governor Eric Holcomb on June 28th, 2019, and for Dean Joanna Millunchick on February 16th, 2023.

Many of the tours were given by Andreas Bueckle and Ezra Engels.



April 11, 2018

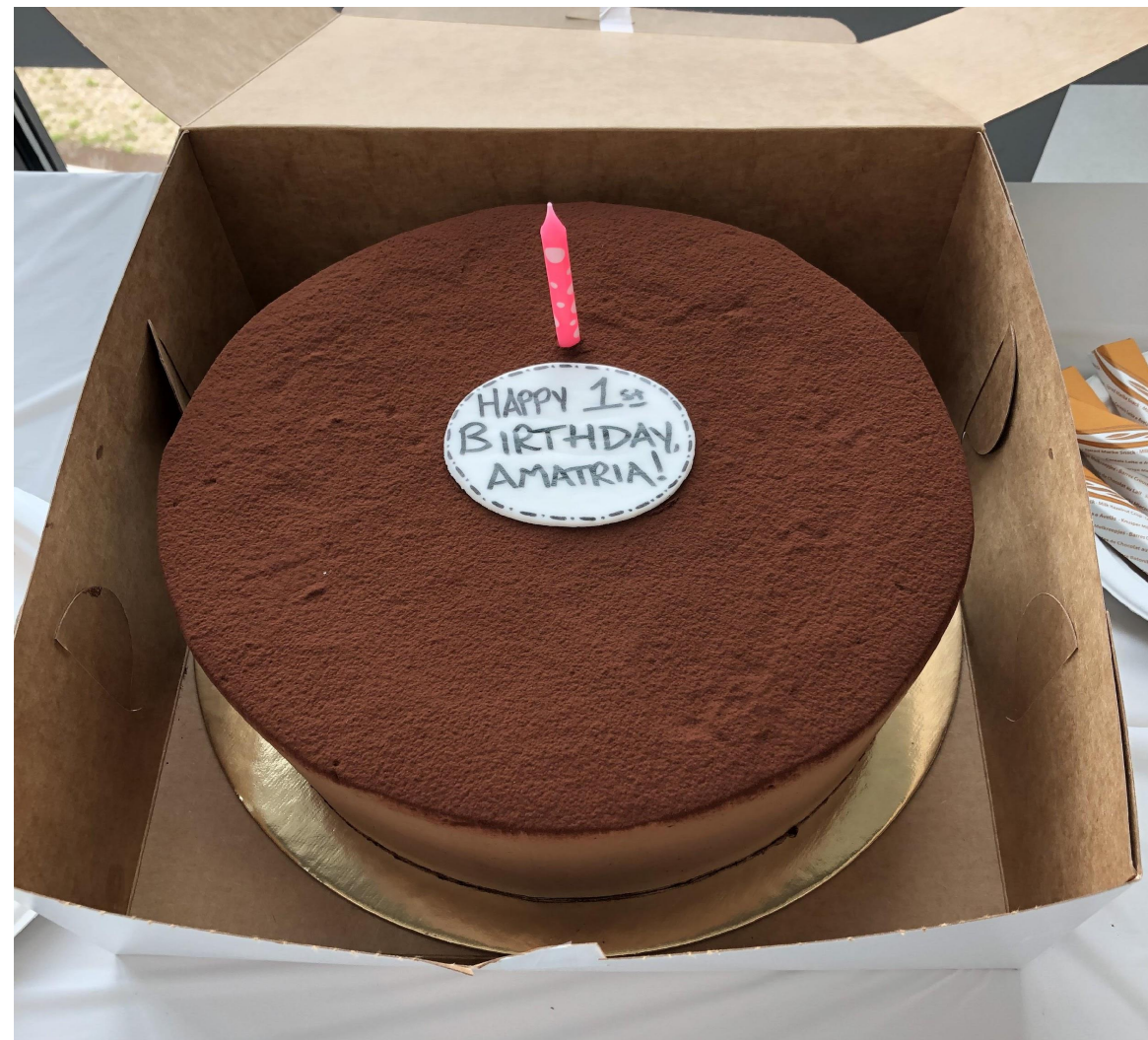
Birth of *Amatria* with 'dad' Philip Beesley and his team



Amatria (2018)

Buddy Hall, SICE, IU

Amatria's 1st Birthday Celebration (04.11.2019)



Internet of Things Wearables in Motion Symposium (04.26.2019)



Visit of Governor Eric Holcomb, IU President Michael McRobbie, Dean Raj Acharya



Tour for Women's Colloquium of IU (10.25.2019)



Amatria Tour for Data Science Online Students (02.28.2020)



Eskenazi Art Museum Visitors (02.26.2020)



Andrew Maple *Amatria* 360 Photo Shoot (12/04/2019)

IU ERI talk/Eric Mathis (LASG) (02/07/2020)

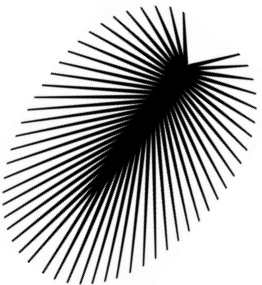
Amatria Vessel Cleaning (01/19/2021 - 3/28/2021)

Amatria tour for Research Experiences for Undergraduates
(06/21/2022)

Amatria tour for Campus Visit for Purdue Polytechnic High School
Students (10/05/2022)

Amatria tour for participants of ISSI Conference (07/03/2023)

Amatria Tour for attendees of the ISSI Conference 2023 (07/05/2023)

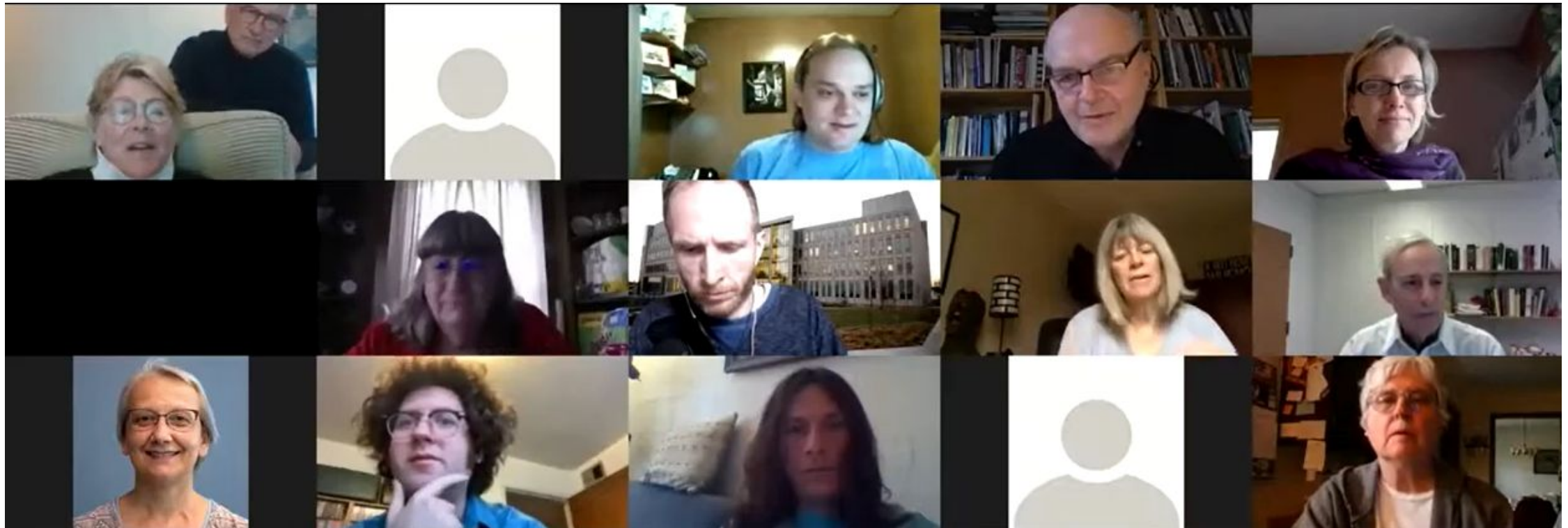




Amatria's 2nd Birthday Celebration (04.13.2020)

Event recording is available on Youtube:

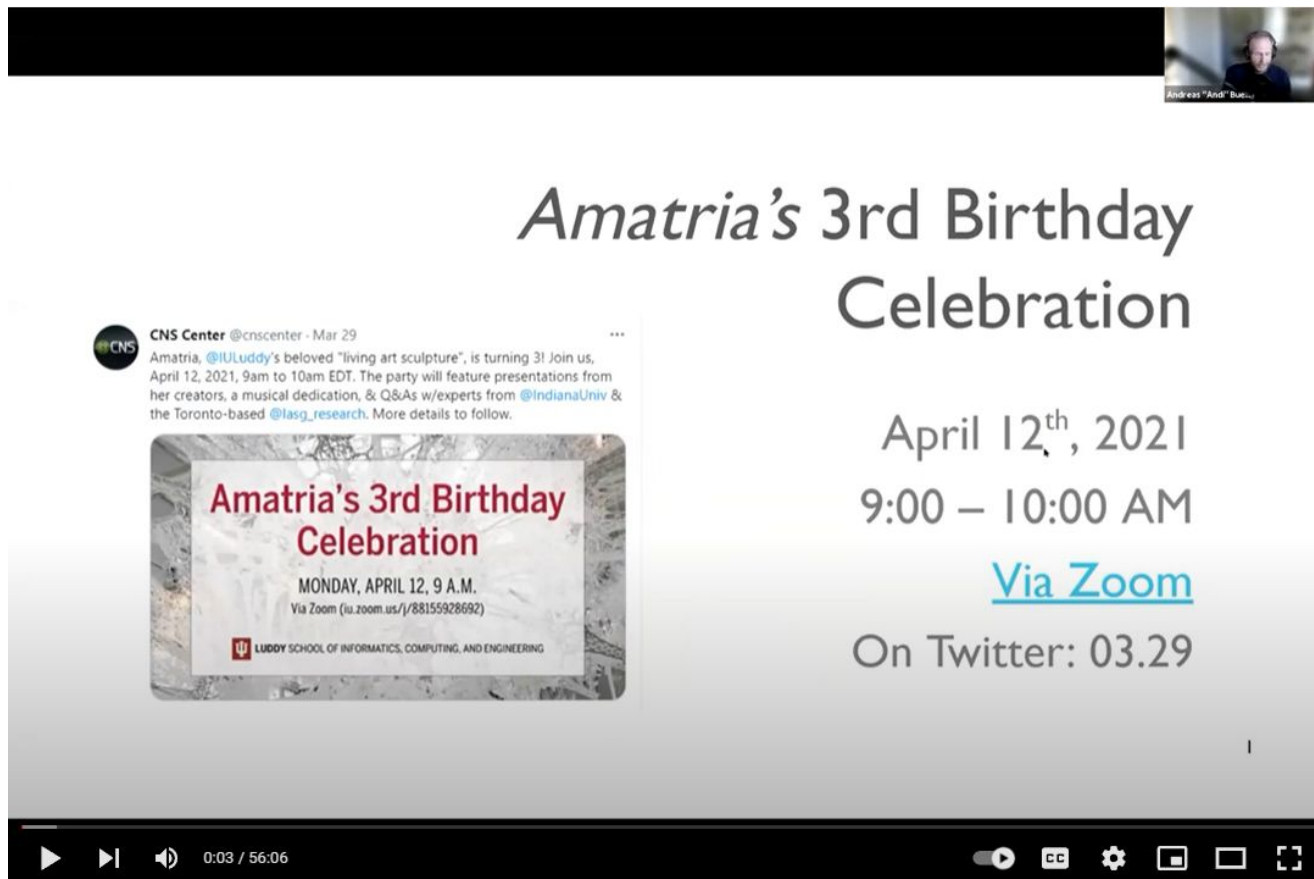
<https://www.youtube.com/watch?v=w3wloIAkjL0&t=332s>



Amatria's 3rd Birthday Celebration (04.12.2021)

Event recording is available on Youtube:

<https://www.youtube.com/watch?v=fR0GdoUimL0>



The image shows a YouTube video player interface. At the top, there is a black bar with a small video thumbnail on the right showing a person. Below this, the video title "Amatria's 3rd Birthday Celebration" is displayed in a large, dark font. To the left of the title, there is a tweet from "CNS Center @cnscenter - Mar 29" about Amatria's 3rd birthday celebration. The tweet text reads: "Amatria, @iUduddy's beloved 'living art sculpture', is turning 3! Join us, April 12, 2021, 9am to 10am EDT. The party will feature presentations from her creators, a musical dedication, & Q&As w/experts from @IndianaUniv & the Toronto-based @lasg_research. More details to follow." Below the tweet is a promotional graphic for the event. The graphic has a textured background and contains the text: "Amatria's 3rd Birthday Celebration", "MONDAY, APRIL 12, 9 A.M.", "Via Zoom (iu.zoom.us/j/88155928692)", and "LUDDY SCHOOL OF INFORMATICS, COMPUTING, AND ENGINEERING". To the right of the graphic, the event details are listed: "April 12th, 2021", "9:00 – 10:00 AM", "Via Zoom" (with a blue link), and "On Twitter: 03.29". At the bottom of the video player, there is a black bar with playback controls, including a play button, a progress bar showing "0:03 / 56:06", and various icons for settings, captions, and full screen.

Amatria's 3rd Birthday Celebration

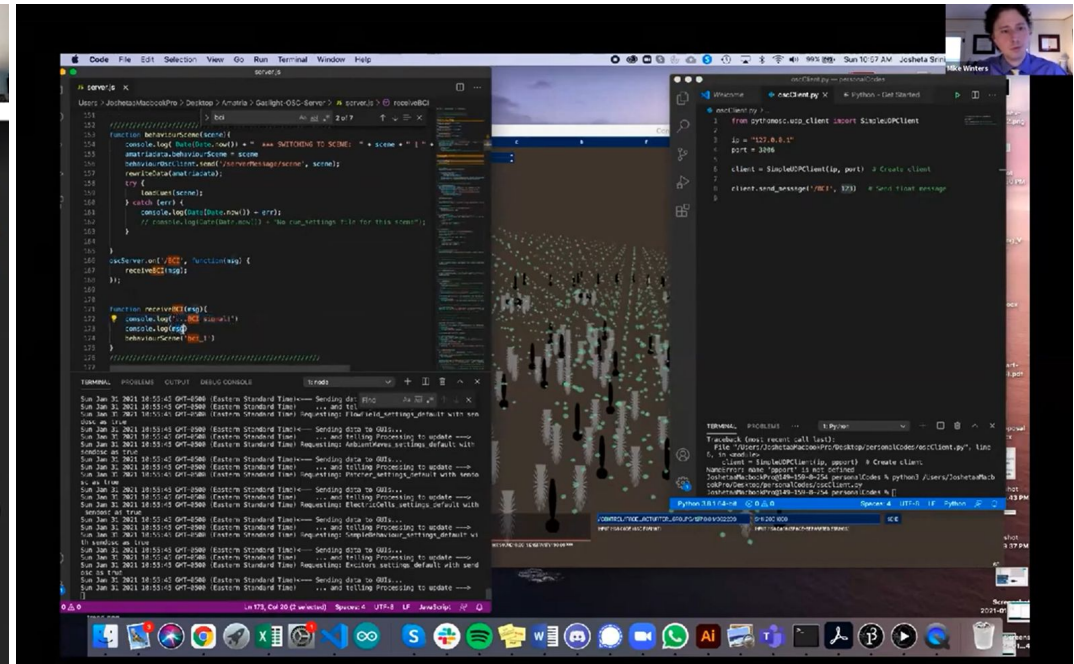
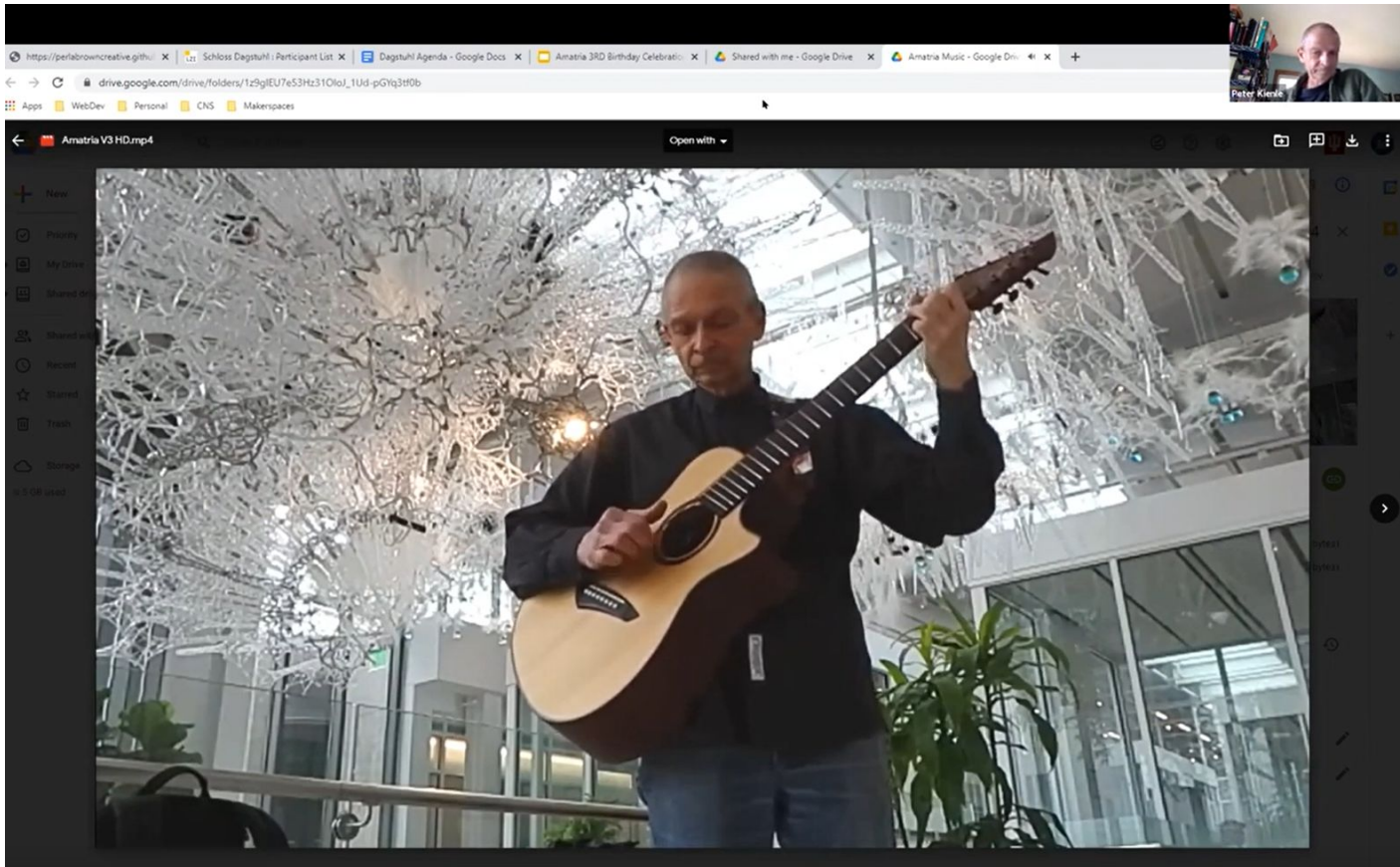
April 12th, 2021
9:00 – 10:00 AM
[Via Zoom](#)
On Twitter: 03.29

CNS Center @cnscenter - Mar 29
Amatria, @iUduddy's beloved "living art sculpture", is turning 3! Join us, April 12, 2021, 9am to 10am EDT. The party will feature presentations from her creators, a musical dedication, & Q&As w/experts from @IndianaUniv & the Toronto-based @lasg_research. More details to follow.

Amatria's 3rd Birthday Celebration
MONDAY, APRIL 12, 9 A.M.
Via Zoom (iu.zoom.us/j/88155928692)
LUDDY SCHOOL OF INFORMATICS, COMPUTING, AND ENGINEERING

April 2021

Amatria's 3rd Birthday Celebration (04.12.2021)

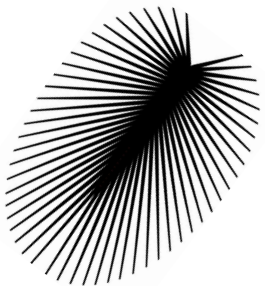


Software update

Amatria is always well taken care of.

In summer 2019, Philip Beesley sent down a team to replace various *Moths* (vibration motors) in *Amatria*'s spar field, which they performed pro bono.

This hardware update was done to prepare for larger software updates to be installed in the near future.



July 2019

Amatria gets a glow-up



August 2021

Amatria tour for Division of Student Affairs



October 2021

Amatria tour for Center of Excellence for Women & Technology's affiliate students



April 2022

First Thursday Festival



June 2022

Amatria tour for Research Experiences for Undergraduates

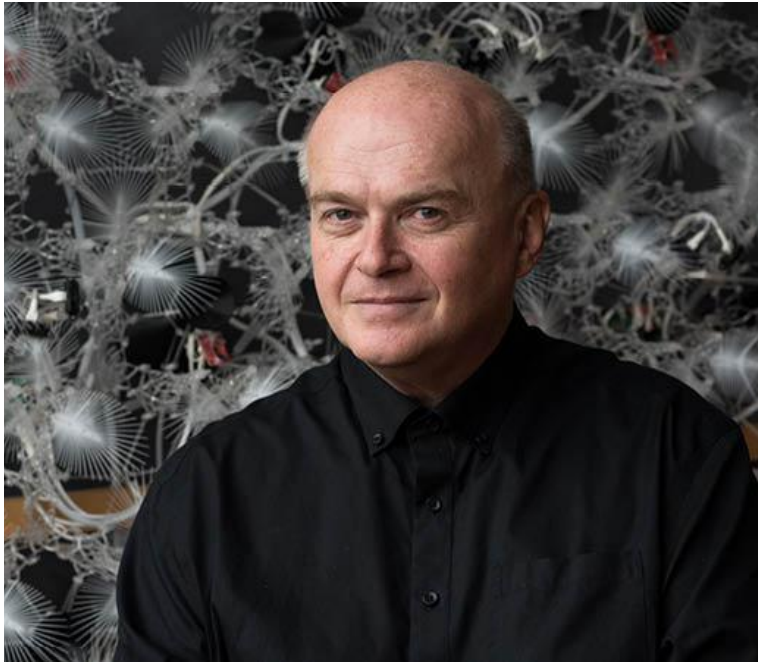


July 2023

ISSI 2023 Conference Tour



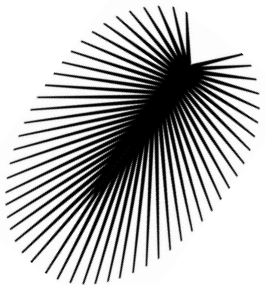
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Dance Performance by Sophia Long



Please maximize your computer audio for better music quality

Tavola:Amatria in 3D,Andreas Bueckle

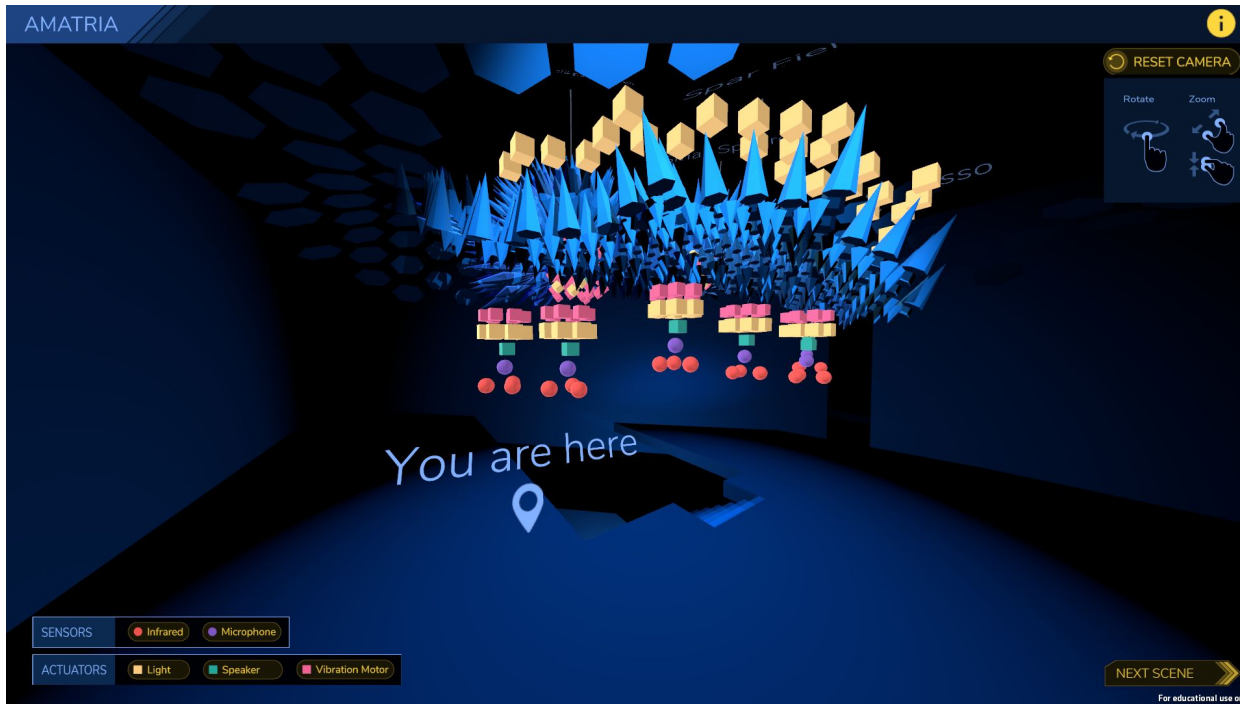


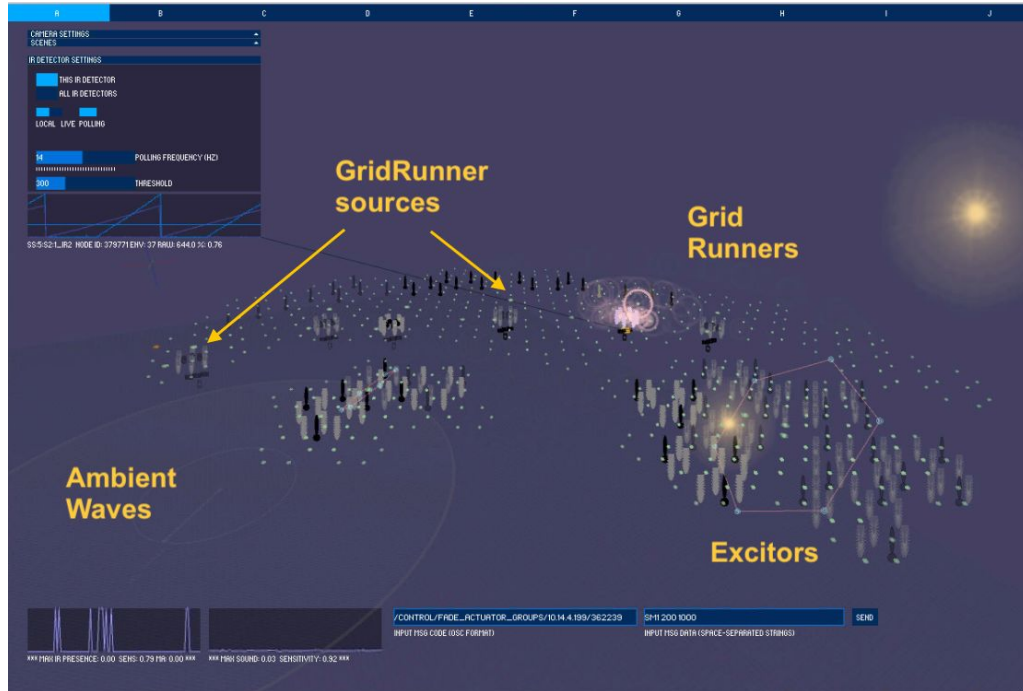
Tavola is a 2018/2019 application developed by Andreas Bueckle (<https://andreas-bueckle.com>). It features a low-poly model of Amatria with a color-coded overlay of all sensors and actuators.

A touch screen camera allows the user to virtually fly through the sculpture and view it from different angles.

An interactive legend allows the user to show/hide sensors and actuators.

The application is deployed to a large touch screen display for today's birthday event!

Processing



- The Amatria behaviour system is based on the ‘influence engine’ model, a suite of layered behaviour algorithms that together create ‘influences’ which affect the realtime behaviour of all actuators in the environment.
- Physics-based ‘Excitor’ particles attracted to orbital ‘attractors’ create a chaotic but smooth flow of movement through the spheres
- Flocking-based ‘GridRunner’ particles released from sources near the IR sensors create local, expanding movement near locations of interaction
- Ambient Waves create coordinated subtle, continuous movement throughout subscribed actuators within the environment
- Loud sudden sounds picked up by the sound detectors create ‘Electric Cell’ activity throughout the sculpture (looks a bit like quick flashing or lightning)
- The sculpture enters 'sleep' mode each night at 8:30pm and wakes at 7:30am (adjustable)



Amrut & Ansh, David Ebbinghouse



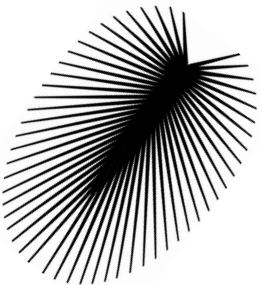
(1' x 1') Plastic bottles, plexiglass, zip ties

(2' x 2') Plastic bottles, plexiglass, mylar, receipt paper spools, LEDs

2022

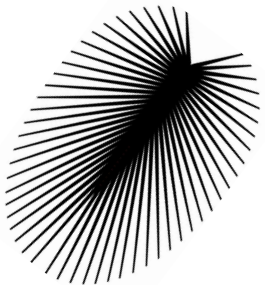
These pieces emerged from Ebbinghouse's experience helping to create Amatria, and incorporate left over plexiglass and mylar from her construction. The circular forms are inspired by cells, and the LEDs in Amrut both draw the viewer's eye and evoke the metabolic process of the cell drawing in and processing nutrients.

Titles were collectively assigned by CNS Center staff, with Ansh meaning 'part' and Amrut meaning 'pure' in Hindi.



Happy *Amatria* Birthday Celebration!

Thank you for joining.



Social Media

Want to learn more about Amatria and what we do? Visit us!



FACEBOOK

<https://www.facebook.com/cnscenter>

X

<https://twitter.com/cnscenter>

Website

<https://cns.iu.edu/amatria.html>

INSTAGRAM

https://www.instagram.com/cns_at_iu/?hl=en

LINKEDIN

<https://www.linkedin.com/company/cns-indiana-university-bloomington/people/>

<https://cns.iu.edu>